

CHAPTER 16

The Neurological Exam

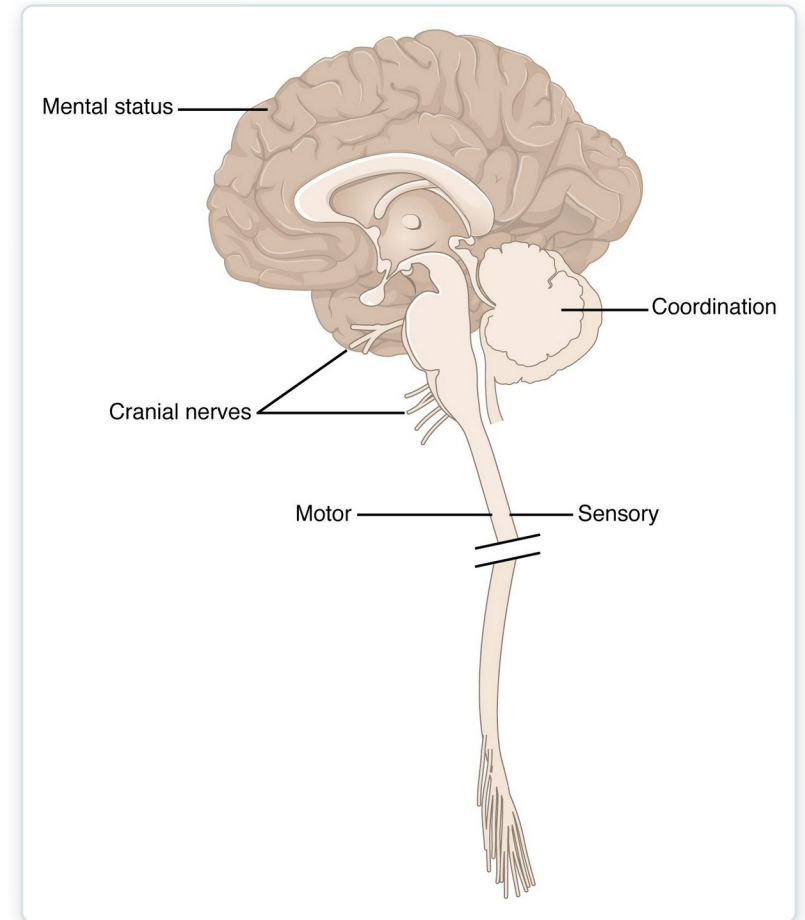
A structured bedside assessment of nervous system function.



Learning Objectives

After studying this chapter, students will be able to:

- 1 Describe the major sections of the neurological exam
- 2 Outline the benefits of rapidly assessing neurological function
- 3 Relate anatomical structures of the nervous system to specific functions
- 4 Diagram the connections of the nervous system to the musculature and integument involved in primary sensorimotor responses



The five parts of the neurological exam.



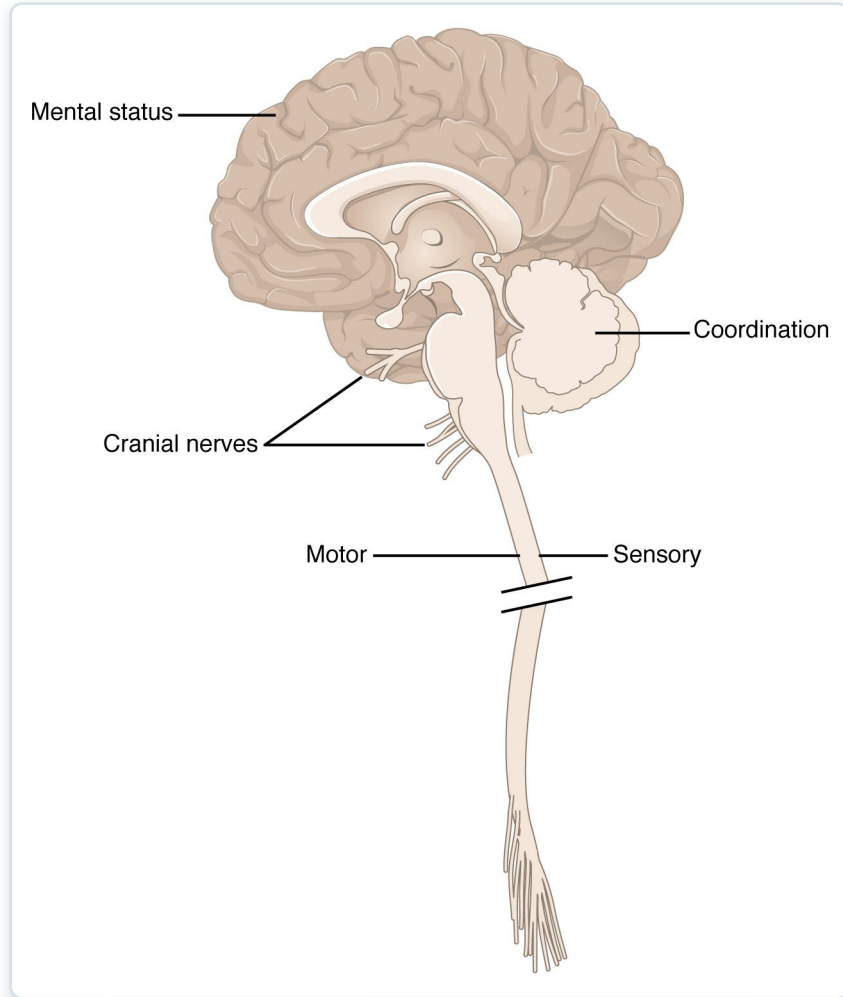
16.1

SECTION

Overview of the Neurological Exam

The neurological exam as a structured tool to localize problems.

The Neurological Exam



The components of the neurological exam

- A structured bedside assessment
- Five major components
- Each tests different brain regions
- Helps localize the problem

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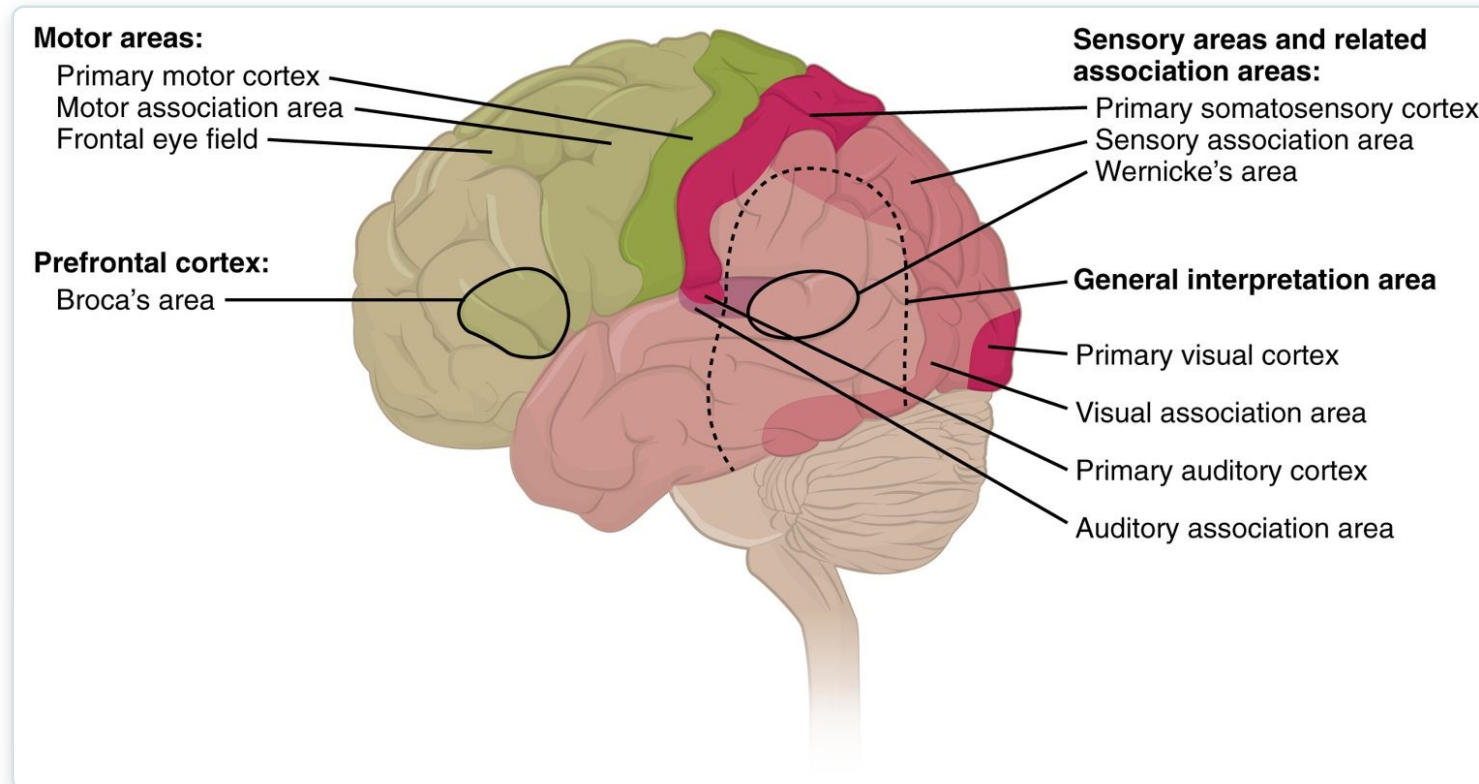
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SECTION

The Mental Status Exam

Testing the higher functions of the cerebral cortex.

Cortical Functional Areas

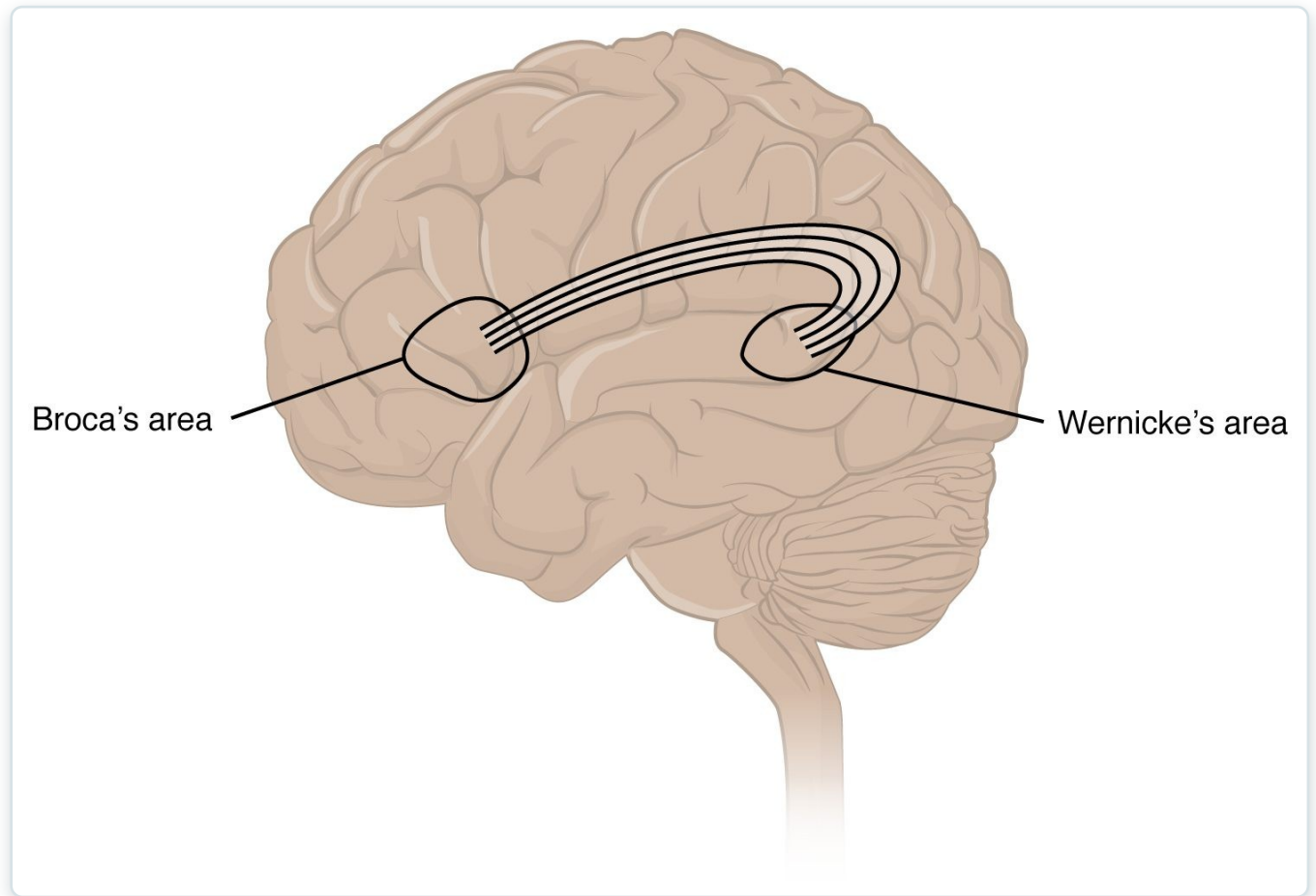


Functional areas of the cerebral cortex

- The cortex has specialized regions
- Motor, sensory, visual, auditory areas
- Association areas integrate information
- Damage gives specific deficits

Language Areas

- Two key language centers
- Broca's area — speech production
- Wernicke's area — comprehension
- Damage causes aphasia



Broca's and Wernicke's language areas

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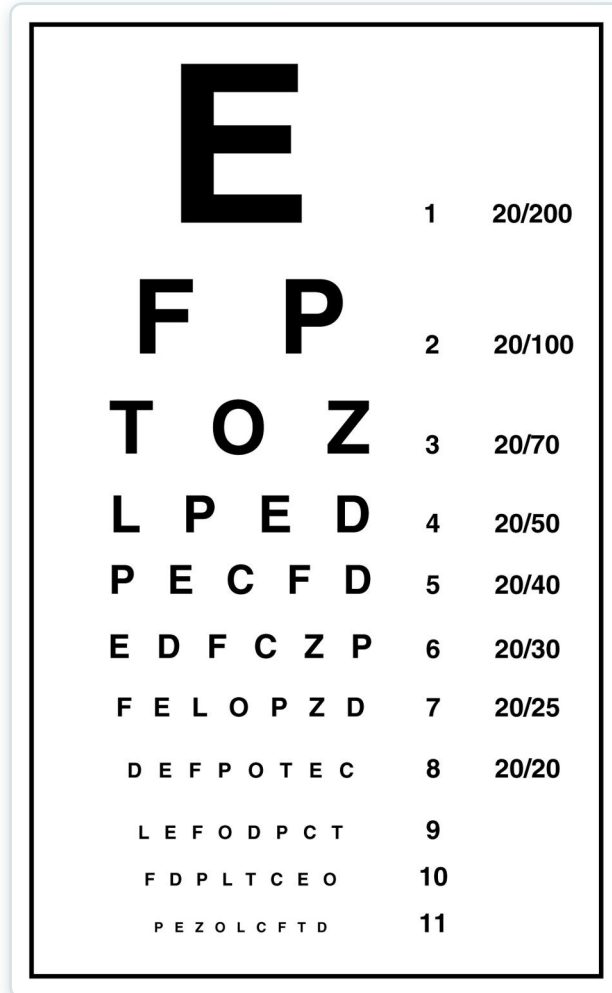
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SECTION

The Cranial Nerve Exam

Testing the twelve cranial nerves and the regions they serve.

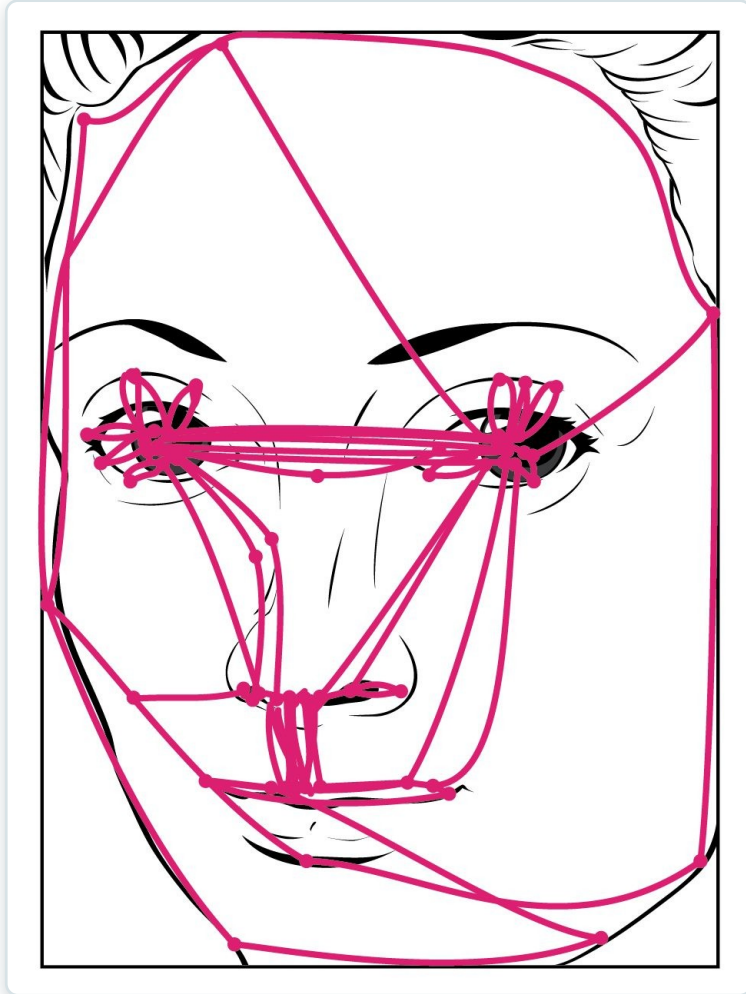
Visual Acuity — CN II



The Snellen eye chart

- The Snellen chart tests vision
- Checks the optic nerve (CN II)
- Each eye is tested separately
- Part of cranial nerve testing

Facial Nerve — CN VII

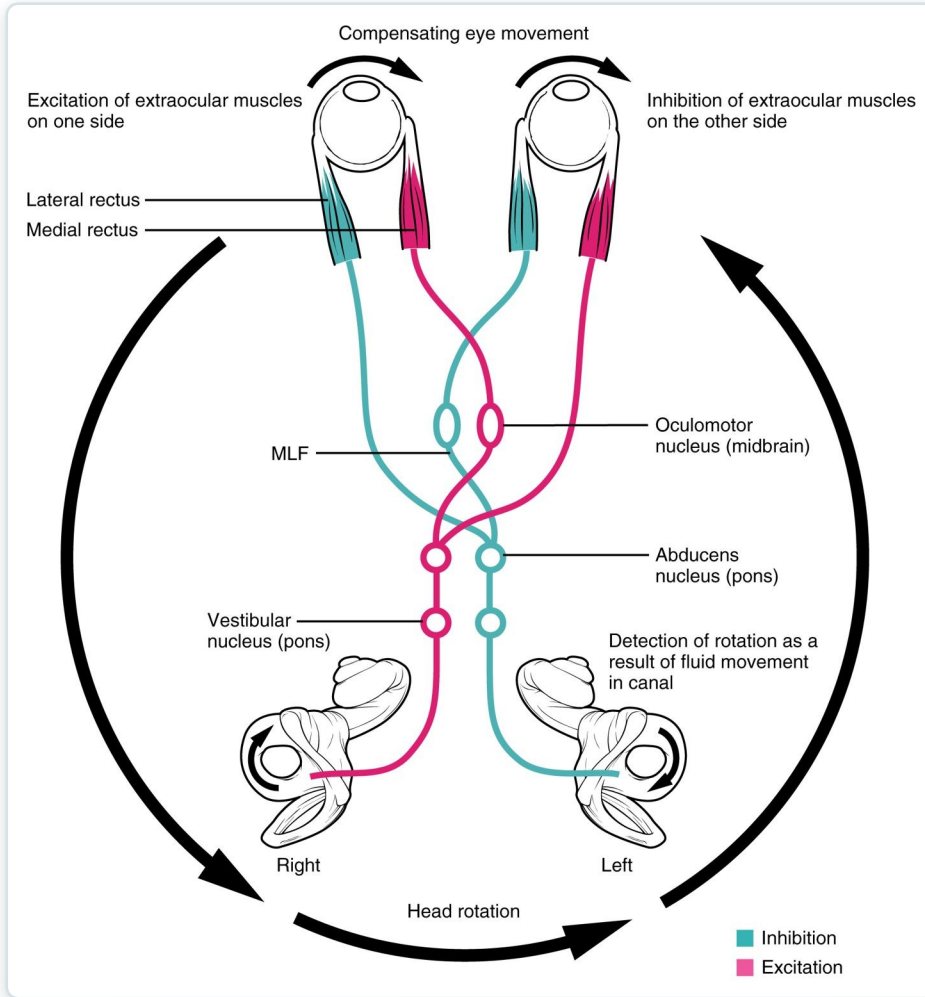


Testing the muscles of facial expression

- Tests the facial expression muscles
- The facial nerve (CN VII)
- Ask the patient to smile and frown
- Weakness localizes the lesion

SECTION 16.3

Eye Movements — CN III, IV, VI

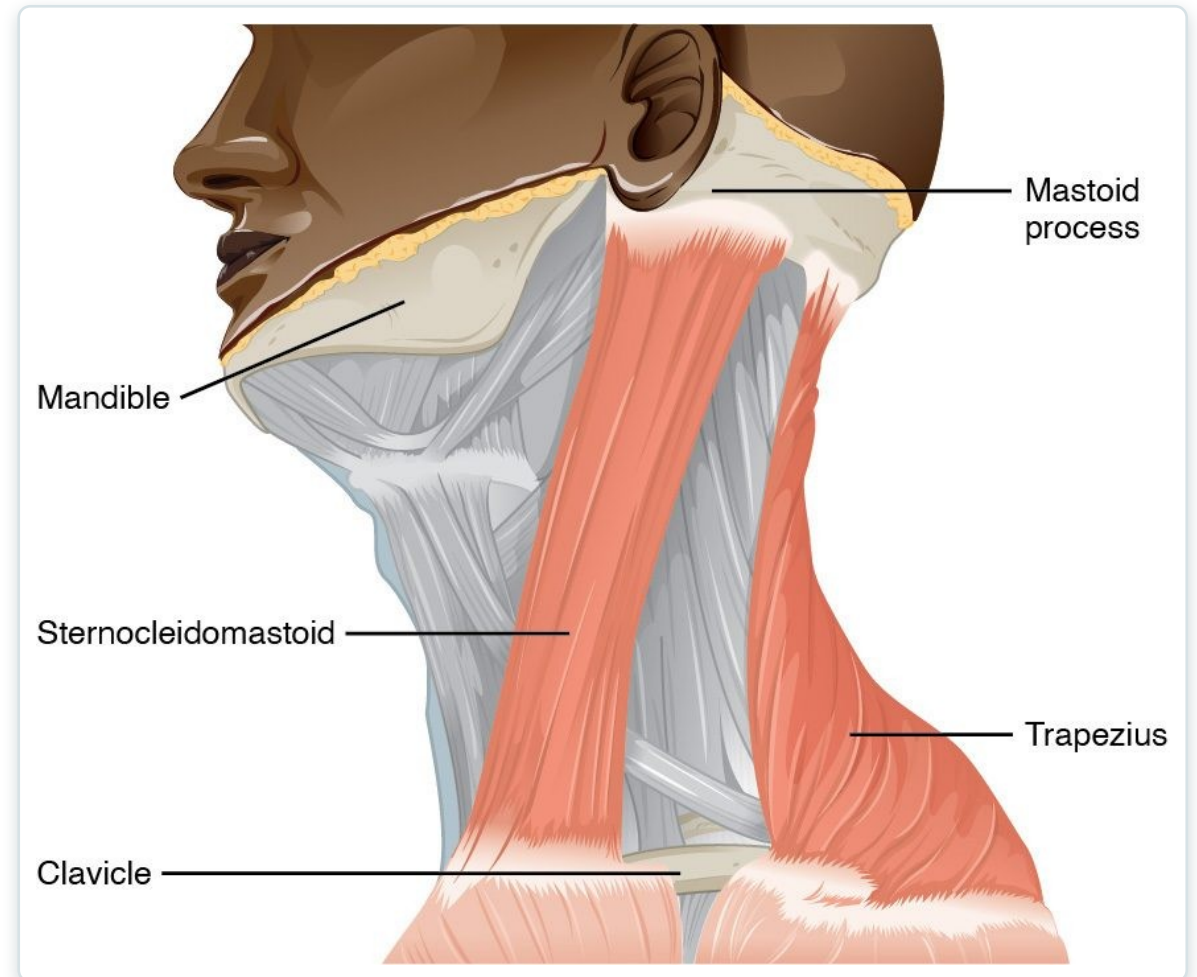


Control of eye movements

- Three nerves move the eyes
- Oculomotor, trochlear, abducens
- Follow a target in all directions
- Detects palsies and gaze problems

Accessory Nerve — CN XI

- Tests neck and shoulder muscles
- The accessory nerve (CN XI)
- Sternocleidomastoid and trapezius
- Shrug and turn-head movements



The sternocleidomastoid and trapezius (CN XI)

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SECTION

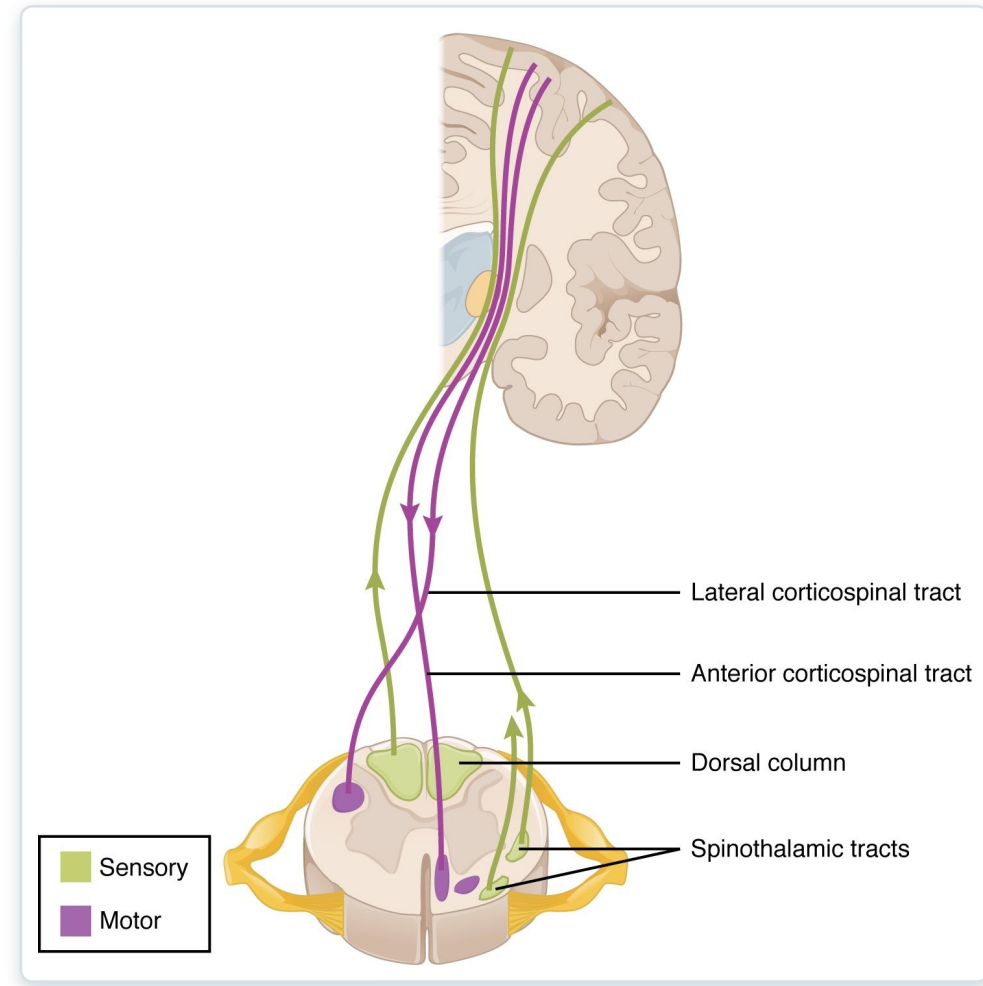
The Sensory and Motor Exams

Testing sensation and voluntary movement throughout the body.

SECTION 16.4

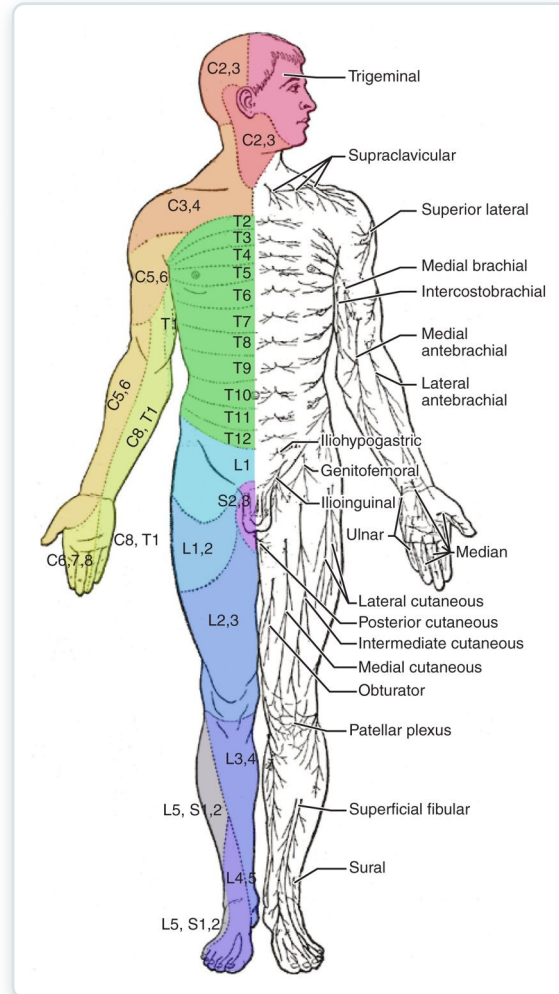
Sensory & Motor Tracts

- Distinct pathways in the spinal cord
- Dorsal column — fine touch, position
- Spinothalamic — pain and temperature
- Corticospinal — voluntary movement



Sensory and motor tracts of the spinal cord

Dermatomes



The dermatome map

- Each spinal nerve maps to a skin area
- These areas are called dermatomes
- Testing them localizes nerve damage
- Guides the level of spinal injury



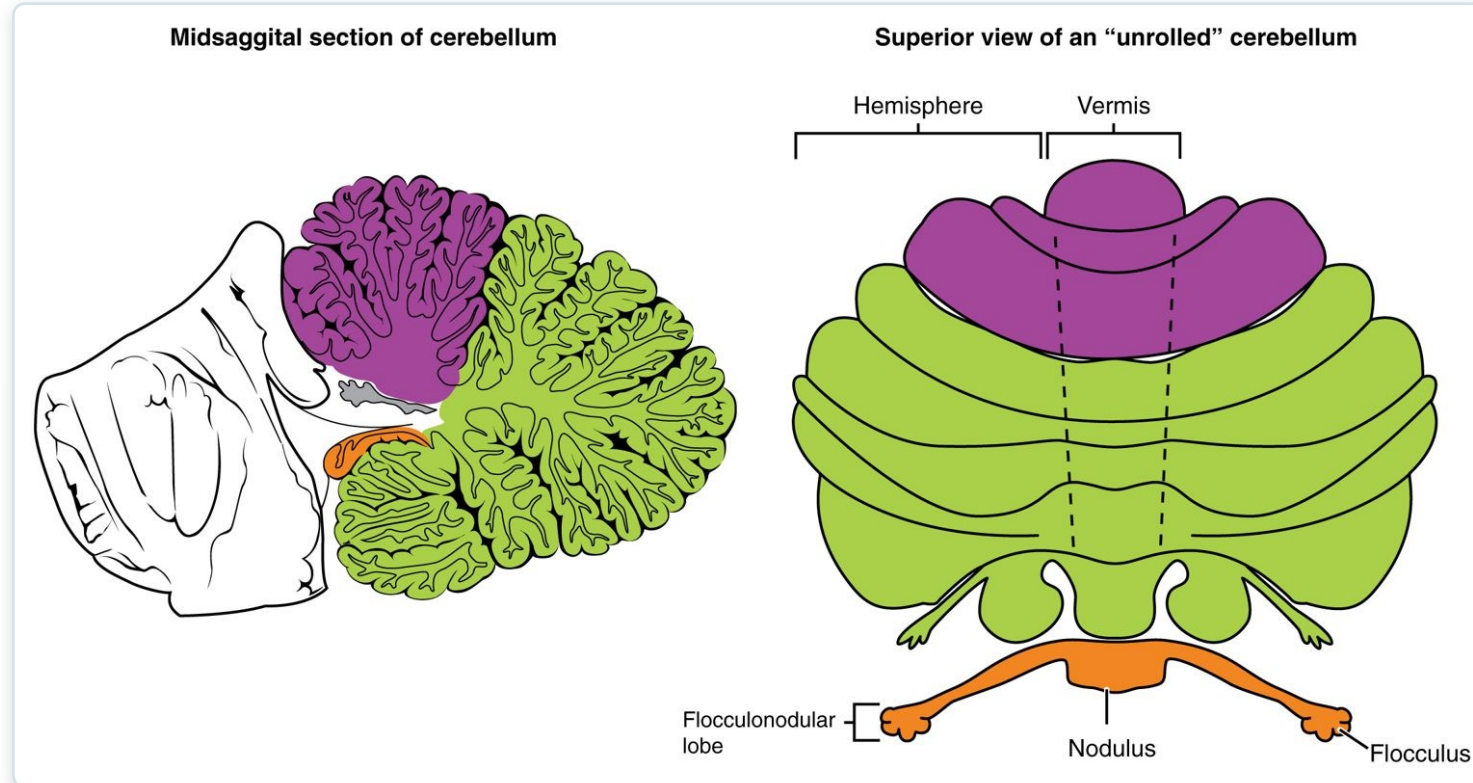
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SECTION

The Coordination and Gait Exams

Testing the cerebellum through coordination and gait.

The Cerebellum



The cerebellum and coordination

- Coordinates movement and balance
- Tested by coordination and gait
- Finger-to-nose, heel-to-shin
- Damage causes ataxia

Key Takeaways

- The neurological exam has five parts that together localize nervous system problems.
- The mental status exam tests cortical functions, including memory and language.
- The cranial nerve exam checks the twelve nerves serving the head and neck.
- The sensory and motor exams test spinal cord pathways and dermatomes.
- The coordination and gait exam assesses the cerebellum.

Why the Neurological Exam Matters in Practical Nursing

Stroke Assessment

Rapid neuro exam findings localize damage and guide treatment.

Aphasia

Speech changes point to specific language-area damage.

Bell's Palsy

Facial weakness sparing the forehead suggests a peripheral cause.

Dermatome Mapping

Sensory loss localizes spinal nerve or cord injury.

Ataxia

Incoordination and unsteady gait suggest cerebellar problems.

Baseline Exams

Serial neuro checks detect changes early in patient care.